Yuchong Pan

Contact Information

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Research Interests

Algorithms, combinatorics, optimization, theoretical computer science, graph theory, operations research.

Education

Massachusetts Institute of Technology

Ph.D., Mathematics, expected May 2027
- Advisor: Michel X. Goemans

University of British Columbia

B.Sc., Computer Science and Mathematics, Honours, with Distinction, Apr 2021
- Thesis: *Optimization Problems on Network Flows with Degree Constraints*
- Advisor: F. Bruce Shepherd

Employment

Microsoft Corporation

Software Engineer Intern, May 2020–Aug 2020
Software Engineer Intern, Jun 2019–Aug 2019
Software Engineer Intern, May 2018–Aug 2018

Sogou, Inc.

Software Engineer Intern, May 2017–Jul 2017

InitialView

Software Engineer Intern, Aug 2016–May 2017

Publications

- Bicriteria approximation algorithms for demand matching (with M. X. Goemans). To appear in *2027 SIAM Symposium on Simplicity in Algorithms (SOSA)*, 2027. Preprint available at `https://arxiv.org/abs/2608.04223`.
- Planarity via spanning tree number: a linear-algebraic criterion (with A. Bu). *SIAM J. Discrete Math.*, 39(2), pp. 728–751, 2025. doi:10.1137/24M1660395.

Manuscripts

- A better-than-3 approximation algorithm for demand matching via knapsack intersection LP and contention resolution (with M. X. Goemans). *arXiv preprint arXiv:2609.17932*, 2026. URL https://arxiv.org/abs/2609.17932.
- An improved integrality-gap reduction for column-restricted packing integer programs. *Unpublished research note; results obtained in 2026.*
- On high-value and high-flow cycles at basic feasible solutions of subtour elimination relaxations for the symmetric and asymmetric traveling salesman problems (with M. X. Goemans). *Manuscript in preparation; results obtained in 2024.*
- A counterexample to box-half-integrality of the intersection of crossing submodular flow systems (with M. X. Goemans). *Unpublished manuscript*, 2023. URL `https://ypan.me/assets/files/counterexample.pdf`.

- Optimization problems on network flows with degree constraints. Undergraduate honours thesis, University of British Columbia, 2021. URL `https://ypan.me/assets/files/honoursthesis.pdf`.

Talks and Presentations

- A better-than-3 approximation algorithm for demand matching. *Theory Lunch*. Computer Science and Artificial Intelligence Laboratory (CSAIL), Massachusetts Institute of Technology. Cambridge, MA. September 29, 2026.
- Bicriteria approximation algorithms for demand matching. *Simple Person's Applied Math Seminar (SPAMS)*. Department of Mathematics, Massachusetts Institute of Technology. Cambridge, MA. September 10, 2026.
- Bicriteria approximation algorithms for demand matching. *The 9th YinzOR Student Conference*. Tepper School of Business, Carnegie Mellon University. Pittsburgh, PA. August 21, 2026.
- Planarity via spanning tree number: a linear-algebraic criterion. *The 2025 AMS New England Graduate Student Conference*. Brown University. Providence, RI. April 19, 2025.
- The tales of planar graphs: the extremal problems, the criterion, and the conjectures. *Meet-and-Think*. Tepper School of Business, Carnegie Mellon University. Pittsburgh, PA. February 19, 2025.
- Fantastic planar graphs and where to find them: a linear-algebraic planarity criterion. *Theory Lunch*. Computer Science and Artificial Intelligence Laboratory (CSAIL), Massachusetts Institute of Technology. Cambridge, MA. February 10, 2025.
- From submodular flows to planar graphs: a planarity criterion inspired by an OR question. *The 7th YinzOR Student Conference*. Tepper School of Business, Carnegie Mellon University. Pittsburgh, PA. August 23, 2024.
- Planarity via spanning tree number: a linear-algebraic criterion. *Algorithms Seminar*. Department of Computer Science, University of British Columbia. Vancouver, BC. May 23, 2024.
- Fantastic spanning trees in planar graphs and where to find them. *Simple Person's Applied Math Seminar (SPAMS)*. Department of Mathematics, Massachusetts Institute of Technology. Cambridge, MA. October 12, 2023.
- Perturbation-stable maximum cuts. *Algorithms Reading Group*. Department of Computer Science, University of British Columbia. Online. June 30, 2020.

Teaching Experience

Massachusetts Institute of Technology

Teaching Assistant (Grader)

18.435J	Quantum Computation, Fall 2026
	Same subject as 2.111J, 6.6410J, 8.370J
18.453/4531	Combinatorial Optimization, Spring 2023
18.085/0851	Computational Science and Engineering I, Fall 2022

Instructor-G (Course Administrator & Recitation Instructor)

18.06	Linear Algebra, Spring 2026

Teaching Assistant (Research Mentor)

18.821	Project Laboratory in Mathematics, Fall 2025

Teaching Assistant (Recitation Instructor)

18.600	Probability and Random Variables, Spring 2025
18.06	Linear Algebra, Spring 2024

University of British Columbia

Teaching Assistant

CPSC 420	Advanced Algorithms Design and Analysis, Spring 2021
CPSC 311	Definition of Programming Languages, Fall 2020
CPSC 421/501	Introduction to Theory of Computing, Fall 2019
CPSC 121	Models of Computation, Fall 2018

Academic Assistant

CPSC 411	Introduction to Compiler Construction, Fall 2019–Spring 2020 *Assisted course redesign, supervised by William J. Bowman.*

Mentoring Experience

Massachusetts Institute of Technology

Directed Reading Program (DRP), Jan 2025–Feb 2025.
- Students: Tiago Oliveira Marques, Eric Shen, Jophy Ye
- Book: *Graphs and Geometry* by László Lovász

Program for Research in Mathematics, Engineering and Science for High School Students (PRIMES-USA, remote), Jan 2024–Dec 2024.
- Students: Christopher K. Bao (Davidson Academy, Reno, NV), Joshua T. Wang (Brookfield Central High School, Brookfield, WI), William Zhao (Dougherty Valley High School, San Ramon, CA)
- Topic: Approximate Minimum Cuts in Graphs and Hypergraphs
- A poster titled *A Faster Approximation Algorithm for the k-Way Cut Problem in Low-Rank Hypergraphs via Multiplicative Weight Update* was presented by Christopher K. Bao in the AMS-PME Undergraduate Student Poster Session at the Joint Mathematics Meetings (JMM) 2025.
- A poster titled *Improved Analysis of the Branching Contraction Algorithm for the Unweighted Hypergraph Minimum k-Way Cut Problem* was presented by Joshua T. Wang in the AMS-PME Undergraduate Student Poster Session at the Joint Mathematics Meetings (JMM) 2025.
- A poster titled *Upper and Lower Bounds on Several Cut Numbers in Hypergraphs via Randomized Contraction* was presented by William Zhao in the AMS-PME Undergraduate Student Poster Session at the Joint Mathematics Meetings (JMM) 2025.
- Christopher K. Bao, Joshua T. Wang, and William Zhao attend Massachusetts Institute of Technology after high school.

Directed Reading Program (DRP), Jan 2024–Feb 2024.
- Students: Jennifer Ai, Neha Pant
- Book: *Algorithmic Game Theory* edited by N. Nisan, T. Roughgarden, É. Tardos and V. V. Vazirani

Program for Research in Mathematics, Engineering and Science for High School Students (PRIMES-USA, remote), Jan 2023–Dec 2023.
- Student: Ruiyang (Raymond) Luo (University High School, Irvine, CA)
- Topic: Cyclic Basis Orderings and Equitability of Matroids
- Ruiyang (Raymond) Luo attends Massachusetts Institute of Technology after high school.

Research Science Institute (RSI), Jun 2023–Aug 2023.
- Student: Alan Bu (Phillips Exeter Academy, Exeter, NH)
- Topic: Connections Between Concatenations of Two Incidence Matrices and Spanning Trees in Planar Graphs
- This project was selected as one of the top 5 out of 100 oral presentations.
- Alan Bu placed 10th (receiving a $40,000 scholarship) in the 83rd Regeneron Science Talent Search (STS) for this project.
- Alan Bu attends Harvard University after high school.

Research Science Institute (RSI), Jun 2023–Aug 2023.
- Student: Deyan Hadzhi-Manich (High School of Mathematics "Dr. Petar Beron", Varna, Bulgaria)
- Topic: Lower and Upper Bounds on Longest Geometrically Increasing Sequences
- Deyan Hadzhi-Manich attends Harvard University after high school.

Directed Reading Program (DRP), Jan 2023–Feb 2023.
- Students: Tianze Jiang, Wayne Zhao
- Book: *Beyond the Worst-Case Analysis of Algorithms* edited by T. Roughgarden

Extended Summer Program in Undergraduate Research (SPUR+), Jun 2022–Aug 2022.
- Students: Evgeniya Artemova, Christina Yu
- Topic: Lower Bounds on the Discrete Newton's Algorithm for the Submodular Line Search Problem

Directed Reading Program (DRP), Jan 2022–Feb 2022.
- Students: Sualeh Asif, Aparna Gupte
- Book: *Randomized Algorithms* by P. Raghavan and R. Motwani

Honors and Awards

- Graduation with Distinction, University of British Columbia, 2021.
- Science Scholar and Dean's Honour List, University of British Columbia, 2018–2021.
- Scholarships at UBC (total CAD$51,550): Work Learn International Undergraduate Research Award, 2021; Stanley M Grant Scholarship in Mathematics, 2019, 2021; Trek Excellence Scholarship, 2018–2020; Faculty of Science International Student Scholarship, 2018–2020; J Fred Muir Memorial Scholarship in Science, 2020; Dean of Science Scholarship, 2018–2019; Marie Kendall Memorial Scholarship in Science, 2018; Joel Harold Marcoe Memorial Scholarship, 2018; Outstanding International Student Award, 2017.
- 27th Place (out of 118 teams), North American Invitational Programming Contest, Open Division (USA + Canada), 2018.
- 11th Place (out of 67 teams), ACM International Collegiate Programming Contest, Pacific Northwest Regional (Division 1), 2017.
- Silver Medal, IOI China Team Selection Competition, China Computer Federation, 2015.
- Bronze Medal, Asia-Pacific Informatics Olympiad, China Computer Federation, 2015.
- First Prize, National Olympiad in Informatics in Provinces (Grades 10–12 Division), China Computer Federation, 2013–2014.
- Second Prize, National Olympiad in Informatics in Provinces (Grades 7–9 Division), China Computer Federation, 2011–2012.

Professional Services

Journal Review: SIAM J. Discrete Math., Math. Program., Oper. Res. Lett.
Conference Review: IPCO 2024, SODA 2022

Relevant Skills

Languages: English, Mandarin
Programming: LaTeX, Python, Julia, C/C++, Java, C#, MATLAB, JavaScript, Go, Racket, Standard ML, MySQL

Last Updated

November 2, 2026